

Ebrahim Hamidi

Department of Civil, Construction, and Environmental Engineering
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Work Authorization: U.S. Permanent Resident

Education

2021-2025	Ph.D. in Civil and Environmental Engineering <i>Department of Civil, Construction and Environment Engineering, The University of Alabama, Tuscaloosa, AL</i> Research Focus: “Compound Flood Assessment Leveraging Remote Sensing Data and Hydrodynamic Simulation in Low-Gradient Coastal Areas”. Advisor: Dr. Hamed Moftakhari
2023-2024	M.Sc. in Civil and Environmental Engineering <i>Department of Civil, Construction and Environment Engineering, The University of Alabama, Tuscaloosa, AL</i> Research Focus: “Integration of Remote Sensing Data and Numerical Simulation for Flood Monitoring Assessment”
2007-2010	M.Sc. in Civil and Environmental Engineering (Hydraulic Structures Major) <i>Department of Civil and Environment Engineering, Shiraz University, Shiraz, Iran</i> Thesis: “Numerical Solution of Water Wave Propagation and Transformation”. Advisor: Dr. Reza Hashemi
2001-2007	B.Sc. in Civil and Structural Engineering <i>Department of Civil Environment Engineering, Persian Gulf University, Bushehr, Iran</i>

Research and Teaching Experiences

May 2025 – Current	Postdoctoral Fellow , North Carolina State University, USA - Contributed to NASA-funded research project. - Performed flood inundation mapping using commercial satellite data (Capella Space, Planet Labs). - Coupled ADCIRC and SFINCS for coastal flood simulations. - Integrated high-resolution satellite data with numerical models to improve flood mapping accuracy.
Jan. 2022 – Apr. 2025	Research Assistant , University of Alabama, USA - Research Assistant: Working on a project funded by NOAA, NSF and USACE. - Mapped flood inundation using free satellite data (Sentinel-1, Sentinel-2, Landsat). - Developed a cloud-based global flood mapping tool. - Simulated compound coastal flooding using Delft3D and HEC-RAS 2D. - Calibrated and validated models with satellite data in data-scarce basins.
Feb. 2023 - Aug. 2023	Course Coordinator , National Water Center Program Summer Institute, Tuscaloosa, USA - Worked with theme leaders, CUAHSI, NWC, and NOAA staff to plan, prepare, and organize the SI and assist the research fellows.
Summer 2022	Research Fellow , National Water Center Program Summer Institute, Tuscaloosa, USA - Developed a coastal-inland coupled BMI for Next Gen NWM.
Spring and Fall 2021	Teacher Assistant , University of Alabama, USA - Graded Water Resources Engineering (CE 378) - Graded Open Channel Flow (CE 471) - Created Hands-on and Taught 2D HEC-RAS
Aug. 2010 – May 2012	Teaching (Part-time), Kavar Scientific Applied School, Iran - Taught Natural hazards on buildings and mitigation measures, Masonry building
Spring 2009	Teaching (Part-time), Pars Institute of Higher Education, Mohr, Fars, Iran - Taught Steel structural design, Concrete technology, English for civil engineers
Spring 2005	Grader , Persian Gulf University, Bushehr, Iran - Graded Steel structural design

Work Experiences

Jun. 2012 - Oct. 2020 (*Full time*) **Senior Civil and Structural Engineer**, Pars Padab Sanaat Consulting Engineers Company, Shiraz, Iran
 Sep. 2007 - May 2012 (*Part time*)

Responsibilities:

- Led engineer and managed engineering projects
- Conducted site inspections and ensured compliance
- Conducted feasibility studies and performed site visits for technical investigations
- Coordinated with architectural, HVAC, mechanical, and piping teams for integrated structural design

Design experiences:

- Designed hydraulic structures, foundations, and retaining walls
- Designed Concrete and Steel support for water transmission pipelines
- Designed industrial steel and concrete structures and buildings
- Designed of municipal masonry, steel and concrete structures
- Designed structural support for historical building

Documents Preparation:

- Prepared calculation books and engineering design specifications
- Prepared economic and technical justification reports
- Prepared engineering calculation books
- Prepared design procedure for supporting steel structures and concrete foundations
- Developed engineering drawings in compliance with industry standards
- Prepared and edited civil and structural specifications and project documentation
- Applied value engineering to improve sustainability and reduce project costs
- Prepared list of structure and building material take off

Research Interests

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|---|--------------------------------|
| - Coastal and Ocean Science | - Remote Sensing Data Analysis |
| - Hydrodynamic, Hydraulic & Hydrologic Modeling | - Geospatial Data Analysis |
| - Machine Learning and Data Science | - Advanced Programming |
| - Natural Hazard Assessment | - Parallel and Cloud Computing |

Journal Publications (In Prees)

1. Peter, Unger, **Hamidi**, Morris, Cohen, Huang (2026). "Flood hazards and socioeconomic vulnerability across Arkansas: A geospatial cyberinfrastructure", Progress in Disaster Science, <https://doi.org/10.1016/j.pdisas.2026.100591>
2. **Hamidi**, Peter, Moftakhari, Moradkhani (2025). "A multi-source remote sensing-based geocommunication tool for global flood monitoring and management", International Journal of Applied Earth Observation and Geoinformation. <https://doi.org/10.1016/j.jag.2025.104701>
3. **Hamidi**, Nazari, Peter, Moftakhari, Moradkhani (2025). "Enhancing Compound Flood Simulation Accuracy and Efficiency in Urbanized Coastal Areas Using Hybrid Meshes and Modified Digital Elevation Model", Sustainable Cities and Society, <https://doi.org/10.1016/j.scs.2025.106184>
4. **Hamidi**, Henrichsen, Sandquist, Zhang, Moftakhari, Ames, Bao, Ferreira, Mandli (2025). "Coupling Coastal and Hydrologic Models Through Next Generation National Water Model Framework", Journal of Hydrologic Engineering, <https://doi.org/10.1061/JHYEFF.HEENG-6343>
5. **Hamidi**, Peter, Muñoz, Moftakhari, Moradkhani (2023). "Fast Flood Extent Monitoring with SAR Change Detection Using Google Earth Engine". IEEE TGRS, <https://doi.org/10.1109/TGRS.2023.3240097>
6. **Hamidi**, Hashemi, Talebbeydokhti, Neill (2012). "Numerical Modelling of the Mild Slope Equation using Localised Differential Quadrature Method", Ocean Engineering, 47, 88–103, <https://doi.org/10.1016/j.oceaneng.2012.03.004>

Under Review Journal Publications

1. **Hamidi**, Peter, Nazari, Moftakhari, Anarde, Moradkhani (2026). "Integrating Multi-Source Remote Sensing and Numerical Simulation Approaches for Enhanced Flood Assessment", <https://dx.doi.org/10.2139/ssrn.5335438>
2. Zand, Moftakhari, **Hamidi**, Moradkhani (2026). "Cloud-Based Machine Learning for Large-Scale Hybrid Flood Mapping Using Remote Sensing and Numerical Simulations".

Conference Presentations

1. Boumis, **Hamidi**, Spencer (2026). “Coastal flood impacts on the built environment in Wells, Maine: Assessing the effectiveness of pool-to-marsh restoration in reducing exposure”, EGU 2026, Vienna, Austria, <https://doi.org/10.5194/egusphere-egu26-131>
2. Zand, Moftakhari, **Hamidi**, Moradkhani (2026). “An Operational Cloud-Based Framework for Rapid Post-Event Flood Mapping Using Remote Sensing and Machine Learning”, 106th AMS Annual Meeting, Houston, TX, USA.
3. **Hamidi**, Anarde, Wang, Smith, McCune, Pavelsky (2025). “High Tide Flood Detection Using Multi-Satellite Remote Sensing Data and In Situ Flood Sensors”, AGU Fall Meeting, New Orleans, LA, USA.
4. McCune, Anarde, Sebastian, Collins, Grimley, **Hamidi**, Hino, Dietrich (2025). “The End of the Road: Present and Future Chronic Flood Risk Along Coastal Roadways and Impacts to Community Livability”, AGU Fall Meeting, New Orleans, LA, USA.
5. Zand, Moftakhari, **Hamidi**, Moradkhani (2025). “GEE-FMF: A Google Earth Engine-Based Machine Learning Framework for Efficient Regional Flood Mapping”, AGU Fall Meeting, New Orleans, LA, USA.
6. **Hamidi**, Anarde, Wang, Smith, Pavelsky (2025). “Using Satellites to Map Flooding Down East”, North Carolina Down East Resilience Network Fall Meeting, Harkers Island, NC, USA.
7. **Hamidi**, et al. (2024). “Enhanced Flood Assessment Through Numerical Simulations and Multi-Source Remote Sensing Data”, AGU Fall Meeting, Washington, D.C., USA.
8. **Hamidi**, et al. (2024). “Advanced Flood Mapping Using Multi-Source Remote Sensing Data and Hydrodynamic Simulations”, AWRA Spring Conference, Tuscaloosa, AL, USA.
9. **Hamidi**, Nazari, Moftakhari, Moradkhani (2023). “Enhancing Compound Coastal Flood Simulation Accuracy and Efficiency with Hybrid Meshes and Corrected Digital Elevation Models”, AGU Fall Meeting, San Francisco, CA, USA, <https://ui.adsabs.harvard.edu/abs/2023AGUFMNH23D0739H/abstract>
10. **Hamidi**, Peter, Moftakhari, Moradkhani (2023). “A Google Earth Engine App for Urgent Flood Mapping”, AGU Fall Meeting, San Francisco, CA, USA, <https://ui.adsabs.harvard.edu/abs/2023AGUFM.H31Y1839H/abstract>
11. Henrichsen, **Hamidi**, Sandquist, Zhang, Ferreira, Mandli (2022). “Coupling Coastal and Hydrologic Models Through the First Coastal Basic Model Interface in the Next Generation National Water Model Framework in Low Gradient Coastal Regions of Galveston Bay, Texas, USA”, AGU Fall Meeting, Chicago, IL, USA, <http://www.hydroshare.org/resource/379b4c8c663c460d87c246641dc5cea2>.
12. **Hamidi**, Peter, Muñoz, Moftakhari, Moradkhani (2022). “Fast Flood Mapping with Synthetic Aperture Radar Data Using Google Earth Engine”, AGU Fall Meeting, Chicago, IL, USA, <https://ui.adsabs.harvard.edu/abs/2022AGUFM.H55M0739H/abstract>
13. **Hamidi**, Munoz, Moftakhari, Moradkhani (2021). “Rapid Coastal Flood Mapping with SAR Data Using Random Forest Technique”, AGU Fall Meeting, New Orleans, LA, USA, <https://ui.adsabs.harvard.edu/abs/2021AGUFM.H335I1138I%2F/abstract>
14. **Hamidi**, Feyli (2015). “Numerical Modelling of Pennes Bioheat Transfer Equation Using Differential Quadrature Method”, The 2nd International Conference on Fluid Flow, Heat and Mass Transfer, Ottawa, Ontario, Canada.

Data and Code Publications

1. **Hamidi**, Peter, Moftakhari, Moradkhani (2024). *Fast Flood Monitoring Tool – FFMT*, A Google Earth Engine App for Fast Flood Monitoring, Hydroshare, <https://doi.org/10.4211/hs.bf66a6cc204d4691abda18833bf68760>
2. Wang and **Hamidi** (2023). *National Water Center Innovators Program - Summer Institute 2023*, CUAHSI Technical Report, Hydroshare, <http://www.hydroshare.org/resource/709de9e2bc5640b281c7f022cecc593d>.
3. **Hamidi**, Peter (2022). *SAR-Based Coastal Flood Extent Estimation Post-Hurricane using Google Earth Engine*, Dataset published on Harvard Dataverse, <https://doi.org/10.7910/DVN/WOTC7E>
4. **Hamidi** (2021). *ArcGIS Script Tool for Flood Extraction from Optical Satellite Data*, <https://github.com/ebrahimhamidi/ArcGIS-Script-Tool-for-Flood-Extraction-from-Optical-Satellite-Data.git>

Awards

- [The University of Alabama Outstanding Dissertation Award](#), February 2026
- [Editor's Choice Article for the Apr. 2025](#) Issue of Journal of Hydrologic Engineering published by ASCE.
- CUAHSI's Hydroinformatics Innovation Fellowship Award, 2023 for [Fast Flood Monitoring Tool](#)
- The National Water Center Innovators Program Award, 2022

Leadership

- [Vice President of Postdoctoral Association](#) at North Carolina State University, 2025-2026
- [Course Coordinator](#) at the National Water Center Innovators Summer Program, 2023
- Led engineering projects at Pars Padab Sanaat Consulting Engineers, 2017-2020

Selected Workshops

- Monitoring Global Terrestrial Surface Water Height using Remote Sensing, NASA's Applied Remote Sensing Training (ARSET), 2025
- Spatial Data Science: The New Frontier in Analytics, 2024, ESRI
- Disaster Assessment Using Synthetic Aperture Radar, NASA's Applied Remote Sensing Training (ARSET), 2022
- Geospatial Storytelling, 2021 BRIGHT online workshop, NCAR
- Hydrodynamic modeling using SCHISM, 2021 NOAA SCHISM online boot camp

Software, Programming & Cloud Platform Skills

- **Programming:** Python, R, MATLAB, C, C++, JavaScript
- **Parallel:** Multithreaded Program (OpenMP), Message-Passing Program (MPI)
- **Software: GIS:** Google Earth Engine, QGIS, ArcGIS pro, ERDAS IMAGINE, SNAP, ArcGIS Drones2Map
- **Hydrodynamics:** Delf3D-FM, 2D HEC-RAS, ADCIRC+SWAN, SFINCS, OpenFoam, National Water Model, GeoClaw
- **Structural:** AutoCad, Sap, Safe, Etabs

Scientific Reviewing

- The International Geoscience and Remote Sensing Symposium (IGARSS), 2026
- Geophysical Research Letters, 2026
- International Journal of Applied Earth Observation and Geoinformation, 2026
- Elsevier Proposal Review, 2026
- Environmental Modeling & Software, 2026
- Water Resources Research, 2025, 2026
- Geomatics, Natural Hazards and Risk, 2025, 2026
- Ocean Modeling, 2025
- Journal of Hydrology, 2025
- International Journal of Digital Earth, 2025
- Remote Sensing Applications Society and Environment, 2025
- CLEAN - Soil, Air, Water, 2025
- Environmental Modelling and Software, 2025
- Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2025, 2024 and 2023
- International Journal of Disaster Risk Science, 2024
- Journal of Hydrologic Engineering, 2024
- Scientific Reports, 2024
- Remote Sensing of Environment, 2023
- Proceedings of the Institution of Mechanical Engineers, Part M: Journal of Engineering for the Maritime Environment, 2014

Online Activities

- [Google Scholar](https://scholar.google.com/citations?user=SQgEMXAAAAAJ&hl=en) [https://scholar.google.com/citations?user=SQgEMXAAAAAJ&hl=en]
- [LinkedIn](https://www.linkedin.com/in/ebrahim-hamidi-30960b69/) [https://www.linkedin.com/in/ebrahim-hamidi-30960b69/]
- [GitHub](https://github.com/ebrahimhamidi) [https://github.com/ebrahimhamidi]
- [Personal Webpage](https://ebrahimhamidi.com/) [https://ebrahimhamidi.com/]

Extracurricular Activities

- Mountain and Rock Climbing, Swimming, Basketball, Skiing, Piano, and Books

References

- **Dr. Hamed Moftakhari**, *Department of Civil, Construction and Environmental Engineering, University of Alabama*, Contact: +1 (205) 348-0239, hmoftakhari@eng.ua.edu
- **Dr. Tamlin M. Pavelsky**, *Department of Earth, Marine and Environmental Sciences, University of North Carolina at Chapel Hill*, pavelsky@unc.edu

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- **Dr. Reza Hashemi**, *Department of Ocean Engineering, University of Rhode Island*, Contact: +1 (401) 874-6217, reza_hashemi@uri.edu